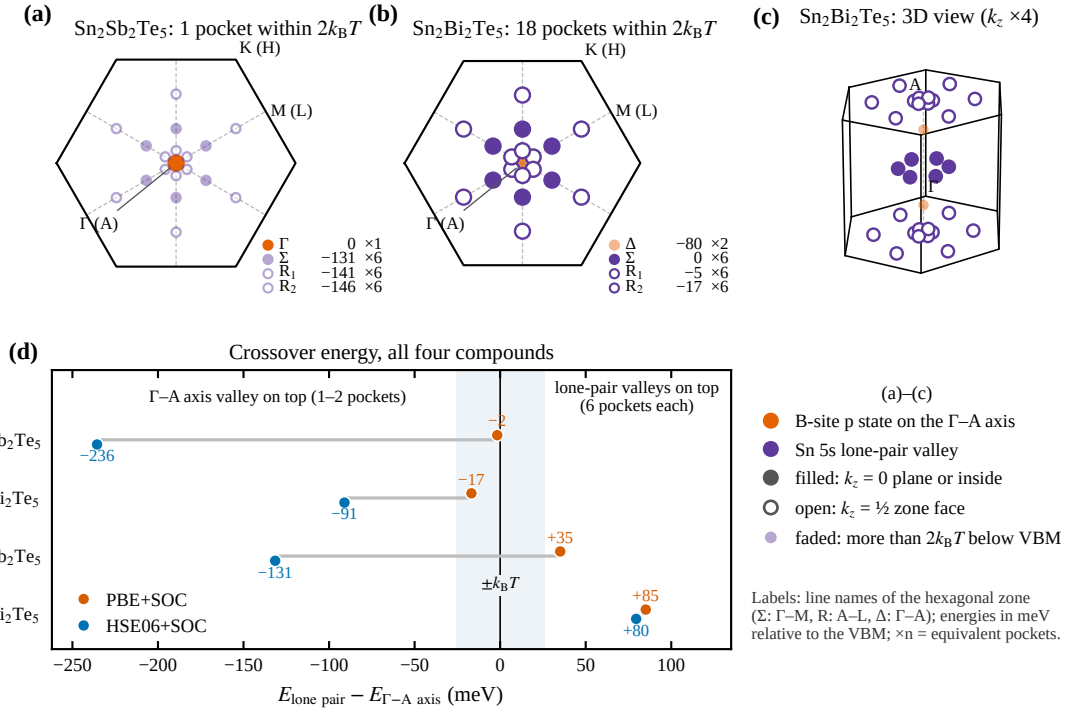
(a)
Sn2Sb2Te5: 1 pocket within 2kBT
K (H)
M (L)
Γ (A)
Γ 0 ×1
Σ −131 ×6
R1 −141 ×6
R2 −146 ×6
(b)
Sn2Bi2Te5: 18 pockets within 2kBT
K (H)
M (L)
Γ (A)
Δ −80 ×2
Σ 0 ×6
R1 −5 ×6
R2 −17 ×6
(c)
Sn2Bi2Te5: 3D view (kz ×4)
A
Γ
(d)
Crossover energy, all four compounds
Γ–A axis valley on top (1–2 pockets)
lone-pair valleys on top
(6 pockets each)
Pb2Sb2Te5
Pb2Bi2Te5
Sn2Sb2Te5
Sn2Bi2Te5
−2
−236
−17
−91
+35
−131
±kBT
+85
+80
PBE+SOC
HSE06+SOC
−250
−200
−150
−100
−50
0
50
100
Elone pair − EΓ−A axis (meV)
(a)–(c)
B-site p state on the Γ–A axis
Sn 5s lone-pair valley
filled: kz = 0 plane or inside
open: kz = ½ zone face
faded: more than 2kBT below VBM
Labels: line names of the hexagonal zone (Σ: Γ–M, R: A–L, Δ: Γ–A); energies in meV relative to the VBM; ×n = equivalent pockets.